

17. To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_JOBS 10
#define MAX_COMMAND 100

typedef enum {
    RUNNING,
    COMPLETED
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    char command[MAX_COMMAND];
    JobState state;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

/* Add a job to the job table */
void addJob(pid_t pid, const char *command) {
    if (job_count >= MAX_JOBS) {
        printf("Job table is full.\n");
        return;
    }

    jobs[job_count].job_id = job_count + 1;
    jobs[job_count].pid = pid;

    strcpy(jobs[job_count].command, command);

    jobs[job_count].state = RUNNING;

    job_count++;
}

/* Update state of each job */
void updateJobStates() {
    int status;

    for (int i = 0; i < job_count; i++) {
        if (jobs[i].state == RUNNING) {
            pid_t result = waitpid(
```

```
                pid_t result = waitpid(
                    jobs[i].pid,
                    &status,
                    WNOHANG
                );

                if (result == jobs[i].pid) {
                    jobs[i].state = COMPLETED;
                }
            }
        }
    }

/* List only active jobs */
void listActiveJobs() {
    updateJobStates();

    printf("\nActive Jobs\n");
    printf("-----\n");
    printf("JOB ID\tPID\tSTATUS\t\tCOMMAND\n");
    printf("-----\n");

    int found = 0;

    for (int i = 0; i < job_count; i++) {
        if (jobs[i].state == RUNNING) {
            printf("[%d]\t%d\tRUNNING\t\t%s\n",
                jobs[i].job_id,
                jobs[i].pid,
                jobs[i].command);

            found = 1;
        }
    }

    if (!found) {
        printf("No active jobs.\n");
    }

    printf("-----\n");
}

/* Display all jobs */
void displayAllJobs() {
    updateJobStates();

    printf("\nAll Jobs\n");
    printf("-----\n");
```

```

/* Display all jobs */
void displayAllJobs() {
    updateJobStates();

    printf("\nAll Jobs\n");
    printf("-----\n");
    printf("JOB ID\tPID\tSTATUS\t\tCOMMAND\n");
    printf("-----\n");

    for (int i = 0; i < job_count; i++) {
        if (jobs[i].state == RUNNING) {
            printf("[%d]\t%d\tRUNNING\t\t%s\n",
                jobs[i].job_id,
                jobs[i].pid,
                jobs[i].command);
        } else {
            printf("[%d]\t%d\tCOMPLETED\t\t%s\n",
                jobs[i].job_id,
                jobs[i].pid,
                jobs[i].command);
        }
    }

    printf("-----\n");
}

/* Retrieve job ID */
void retrieveJobID(int index) {
    if (index < 0 || index >= job_count) {
        printf("Invalid job number.\n");
        return;
    }

    printf("Job ID: [%d]\n",
        jobs[index].job_id);
}

/* Create a background job */
void launchJob(const char *command, int seconds) {
    pid_t pid = fork();

    if (pid < 0) {
        |

```

```

GNU nano 8.7.1
void launchJob(const char *command, int seconds) {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return;
    }

    if (pid == 0) {
        /* Simulate a background task */
        sleep(seconds);

        exit(0);
    }

    /* Parent stores job information */
    addJob(pid, command);

    printf("[Job %d] Started with PID %d\n",
        jobs[job_count - 1].job_id,
        pid);
}

int main() {
    printf("Job Listing System\n");
    printf("-----\n");

    /* Create two test jobs */
    launchJob("sleep 5", 5);
    launchJob("sleep 8", 8);

    /* Display active jobs */
    listActiveJobs();

    /* Retrieve first job ID */
    printf("\nRetrieving Job ID:\n");
    retrieveJobID(0);

    /* Wait for first job to finish */
    printf("\nWaiting 6 seconds...\n");
    sleep(6);

    /* Display updated status */
    printf("\nUpdated Job Listing:\n");
    displayAllJobs();

    /* Display only active jobs */
    printf("\nActive Job Listing:\n");

```

```

displayAllJobs();

/* Display only active jobs */
printf("\nActive Job Listing:\n");
listActiveJobs();

/* Wait for second job */
sleep(3);

/* Final status */
printf("\nFinal Job Status:\n");
displayAllJobs();

return 0;
}

```

Result:

```
abhinav@Abhinav: ~/SKILL$ nano Skill17a.c
abhinav@Abhinav: ~/SKILL$ gcc Skill17a.c -o s17a
Job Listing System
[Job 1] Started with PID 3056
[Job 2] Started with PID 3057

Active Jobs
-----
JOB ID  PID      STATUS      COMMAND
-----
[1]     3056     RUNNING     sleep 5
[2]     3057     RUNNING     sleep 8
-----

Retrieving Job ID:
Job ID: [1]

Waiting 6 seconds...

Updated Job Listing:

All Jobs
-----
JOB ID  PID      STATUS      COMMAND
-----
[1]     3056     COMPLETED   sleep 5
[2]     3057     RUNNING      sleep 8
-----

Active Job Listing:

Active Jobs
-----
JOB ID  PID      STATUS      COMMAND
-----
[2]     3057     RUNNING      sleep 8
-----

Final Job Status:

All Jobs
-----
JOB ID  PID      STATUS      COMMAND
-----
[1]     3056     COMPLETED   sleep 5
[2]     3057     COMPLETED   sleep 8
-----
abhinav@Abhinav: ~/SKILL$ |
```

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <signal.h>
#include <string.h>
#include <termios.h>

#define MAX_JOBS 5
#define MAX_COMMAND 100

typedef enum {
    RUNNING,
    STOPPED,
    COMPLETED
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    char command[MAX_COMMAND];
    JobState state;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

/* Add a job */
void addJob(pid_t pid, const char *command) {
    if (job_count >= MAX_JOBS) {
        printf("Job table is full.\n");
        return;
    }

    jobs[job_count].job_id = job_count + 1;
    jobs[job_count].pid = pid;
    strcpy(jobs[job_count].command, command);
    jobs[job_count].state = RUNNING;
    job_count++;
}

/* Display jobs */
void displayJobs() {
    printf("\nJob Table\n");
    printf("-----\n");
    printf("ID\tPID\tSTATE\t\tCOMMAND\n");
    printf("-----\n");
}
```

```

GNU nano 8.7.1
printf("-----\n");

for (int i = 0; i < job_count; i++) {
    char *state;

    if (jobs[i].state == RUNNING)
        state = "RUNNING";
    else if (jobs[i].state == STOPPED)
        state = "STOPPED";
    else
        state = "COMPLETED";

    printf("[%d]\t%d\t%s\t\t%s\n",
           jobs[i].job_id,
           jobs[i].pid,
           state,
           jobs[i].command);
}

printf("-----\n");
}

/* Find job by job ID */
int findJob(int job_id) {
    for (int i = 0; i < job_count; i++) {
        if (jobs[i].job_id == job_id)
            return i;
    }

    return -1;
}

/* Switch job to foreground */
void foregroundJob(int job_id) {
    int index = findJob(job_id);

    if (index == -1) {
        printf("Job [%d] not found.\n", job_id);
        return;
    }

    Job *job = &jobs[index];

    printf("\nTarget Job: [%d]\n", job->job_id);
    printf("Command: %s\n", job->command);

    /*

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/*
 * Transfer terminal control to the job.
 */
if (tcsetpgrp(STDIN_FILENO, job->pid) == -1) {
    perror("tcsetpgrp");
}

printf("Terminal control transferred to job [%d].\n",
       job->job_id);

/*
 * Resume stopped process.
 */
if (job->state == STOPPED) {
    if (kill(job->pid, SIGCONT) == -1) {
        perror("SIGCONT");
    }

    job->state = RUNNING;
    printf("Job [%d] resumed.\n",
           job->job_id);
}

/*
 * Wait for foreground process.
 */
int status;
waitpid(job->pid, &status, WUNTRACED);

/*
 * Update job state.
 */
if (WIFSTOPPED(status)) {
    job->state = STOPPED;
    printf("Job [%d] stopped.\n",
           job->job_id);
}

else if (WIFEXITED(status) ||
         WIFSIGNALED(status)) {
    job->state = COMPLETED;
    printf("Job [%d] completed.\n",
           job->job_id);
}

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        job->job_id);
    }

    /*
     * Give terminal back to shell.
     */
    if (tcsetpgrp(STDIN_FILENO, getpgrp()) == -1) {
        perror("tcsetpgrp");
    }

    printf("Terminal control returned to shell.\n");
}

/* Create a test foreground/background job */
void createJob() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return;
    }

    if (pid == 0) {
        /*
         * Create a new process group.
         */
        setpgid(0, 0);

        /*
         * Simulate a job.
         */
        printf("Child process started.\n");

        sleep(5);

        printf("Child process finished.\n");

        exit(0);
    }

    /*
     * Parent creates process group.
     */
    setpgid(pid, pid);

    addJob(pid, "sleep 5");

    printf("Job [%d] created with PID %d.\n",
        jobs[job_count - 1].job_id,
        pid);
}

```

```

/*
 * Parent creates process group.
 */
setpgid(pid, pid);

addJob(pid, "sleep 5");

printf("Job [%d] created with PID %d.\n",
    jobs[job_count - 1].job_id,
    pid);
}

int main() {
    printf("Foreground Job Switching System\n");
    printf("-----\n");

    /*
     * Create test job.
     */
    createJob();

    /*
     * Give child a moment to start.
     */
    sleep(1);

    displayJobs();

    /*
     * Switch job to foreground.
     */
    printf("\nSwitching Job [1] to foreground...\n");
    foregroundJob(1);

    /*
     * Display final state.
     */
    printf("\nFinal Job State:\n");

    displayJobs();

    printf("\nForeground switching completed successfully.\n");

    return 0;
}

```

Result:

```
abhinav@Abhinav:~/SKILL$ nano Skill17b.c
abhinav@Abhinav:~/SKILL$ gcc Skill17b.c -o s17b
abhinav@Abhinav:~/SKILL$ ./s17b
Foreground Job Switching System
-----
Job [1] created with PID 3115.
Child process started.

Job Table
-----
ID      PID      STATE      COMMAND
-----
[1]     3115     RUNNING   sleep 5
-----

Switching Job [1] to foreground...

Target Job: [1]
Command: sleep 5
Terminal control transferred to job [1].
Child process finished.
Job [1] completed.

[1]+  Stopped                  ./s17b
abhinav@Abhinav:~/SKILL$ |
```